

Predictive E-Commerce Analytics & RFM Segmentation

Data Science Portfolio | Julio Cesar Cortez Machado

1. Executive Summary

Developed a comprehensive data ecosystem for a UK-based retailer, processing over **500,000 transactions**. The project covers the end-to-end data lifecycle: from ETL and SQL automation to advanced Machine Learning for customer retention.

Tech Stack:

Python (Pandas)

SQL Server

Scikit-Learn

SQLAlchemy

Power BI

2. Data Pipeline & ETL (Extraction, Transformation, Load)

Using `limpieza.py` and `procesamiento.py`, the workflow was automated:

- **Ingestion:** Seamless integration with SQL Server using SQLAlchemy.
- **Data Cleaning:** Handled missing CustomerIDs and filtered out invalid transactions (negative quantities/prices).
- **Feature Engineering:** Calculated 'Total_Sales' and optimized schemas for BI reporting.

3. Customer Intelligence (RFM Modeling)

Implemented an **RFM (Recency, Frequency, Monetary)** framework in `modelo_rfm.py` to segment the database into 10 actionable groups:

- **Champions & Loyal Customers:** High-value users focused on rewards.
- **At Risk & Hibernating:** Identifying churn candidates for urgent re-engagement campaigns.

4. Predictive Modeling (Machine Learning)

Applied a **Random Forest Classifier** in `prediccion_fuga.py` to anticipate customer churn:

Component	Technical Detail
Algorithm	Random Forest (200 Estimators)
Target Variable	Churn (Inactivity > 90 days)
Optimization	Balanced Class Weighting for skewed datasets

5. Business Impact

By shifting from descriptive to predictive analytics, this project enables businesses to optimize marketing budgets by 20-30% through targeted retention of high-risk, high-value segments.

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